Artwork Document Information

Product : T BUTALINE SYRUP 90ML [FRONT]
SAP No. :
Date : 24 March 2025

Artwork Type [Please Tick /]

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Remarks

1. Font : Adobe Garamond Pro Regular (Spacing -8pt / Size - 6.5pt)
2. Dimension : 170mm(w) x 198mm(h)

Colour

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Butaline Syrup

1.5mg/5ml

DESCRIPTION

A clear colourless fruity flavoured syrup.
Each 5 ml contains Terbutaline Sulphate 1.5 mg

INDICATIONS

Terbutaline is indicated for the symptomatic treatment of bronchial asthma. It is also indicated for the treatment of reversible airways obstruction associated with bronchitis, pulmonary emphysema, bronchiectasis and other chronic obstructive pulmonary diseases. Terbutaline also decreases uterine contractility and may be used to arrest premature labour.

PHARMACODYNAMICS

Terbutaline is a sympathomimetic agent which mimics the actions produced by stimulation of postganglionic sympathetic or adrenergic nerves, including stimulation of the heart and central nervous systems, vasoconstriction of blood vessels supplying skin and mucous membranes, dilatation of the bronchi and blood vessels supplying skeletal muscle, and modulation of metabolism. Terbutaline is a beta-2 receptor agonist. Its bronchodilator effect is due to stimulation of beta-2 receptors in the lungs to relax bronchial smooth muscle, thereby relieving bronchospasm, increasing vital capacity, decreasing residual volume, and reducing airway resistance. This actions is believed to result from increased production of cyclic adenosine 3', 5'-monophosphate (cAMP) caused by activation of the enzyme adenyl cyclase, the enzyme that catalyses the conversion of adenosine triphosphate (ATP) to cAMP. Increased cAMP concentrations, in addition to relaxing bronchial smooth muscle, inhibit release of mediators of immediate hypersensitivity from cells, especially mast cells. Its bronchodilating activity makes it useful in respiratory disorders and has a prime role in the management of disease like asthma and chronic obstructive pulmonary disease. It also relaxes uterine muscle, thereby inhibiting uterine contractions and is used in premature labour to delay childbirth. Other effects include beta-2 receptor mediated decrease in gastrointestinal tone, and increases in renin secretion, hepatic glycogenolysis/gluneogenesis, and pancreatic beta cell secretion.

PHARMACOKINETICS

Terbutaline is variably absorbed (30%-70%) from the gastrointestinal tract. Food reduces the bioavailability of terbutaline by one-third. It is subject to fairly extensive first-pass metabolism by sulphate (and some glucuronide) conjugation in the liver and the gut wall. Onset of action after oral administration is 60-120 minutes while peak effects occur within 2-3 hours. It is active for 4-8 hours following oral administration. It is primarily excreted in the urine partly as the inactive conjugates and partly as unchanged terbutaline. There is some placental transfer.

DOSAGE

Adults
Bronchodilator:
2.5-5 mg 3 times daily.
Labour (premature) inhibitor-maintenance:
2.5 mg 4-6 hourly OR 5 mg 3 times daily until term.
Usual adult prescribing limit: 15 mg daily.
Children
Children 12-15 years: 2.5 mg 3 times daily.
Children up to 12 years: Dosage has not been established.

SIDE EFFECTS

Terbutaline may produce a wide range of adverse effects, most of which mimic the result of excessive stimulation of the sympathetic nervous system. These effects are mediated via the various types of adrenergic receptors. Frequent side effects reported with terbutaline include nervousness or restlessness, trembling, anxiety and irritability. Dizziness or lightheadedness, drowsiness, tachycardia, headache, increased sweating, hypersalivation, increase in blood pressure, muscle cramps or twitching, weakness, pounding heartbeat, difficulty in sleeping, nausea, vomiting and unusual taste have been reported. The rise in blood pressure may produce cerebral haemorrhage and pulmonary edema. Pulmonary edema is particularly important when terbutaline is used for delaying premature labour. Rarely, chest discomfort or chest pain, irregular heartbeat and paradoxical bronchospasm (increase in wheezing or difficulty in breathing) also occur. Altered metabolism including changes in blood sugar concentrations may occur.

CONTRAINDICATIONS

Terbutaline is contraindicated in patients with hypersensitivity to any of the sympathomimetics. It should be avoided in patients undergoing anaesthesia with cyclosporine, halothane or other halogenated anaesthetics, as they may induce ventricular fibrillation. An increased risk of arrhythmia may also occur if terbutaline is given to patients receiving cardiac glycosides, quinidine, or tricyclic antidepresants.

PRECAUTIONS

Terbutaline should be used with caution in patients who be particularly susceptible to its effects, especially those with hyperthyroidism. Great care is also needed in patients with cardiovascular diseases such as ischaemic heart disease, arrhythmia or tachycardia, occlusive vascular disorders including arteriosclerosis, hypertension, or aneurysms. Anginal pain may be precipitated in patients with angina pectoris. Care is also needed when terbutaline is given to patients with diabetes mellitus, closed-angle glaucoma, phaechromocytoma and history of seizures.
Serum potassium concentrations may be decreased, possibly through intracellular shunting, when terbutaline is given intravenously or by nebulisation or in higher-than-recommended doses; decrease is usually transient and supplementation is not required.
Tocolysis : Serious adverse reactions including death have been reported after administration of terbutaline/sabutamol to women labor. In the mother, these include increased heart rate, transient hyperglycaemia, hypokalaemia, cardiac arrhythmias, pulmonary oedema and myocardial ischaemia. Increased fetal heart rate and neonatal hypoglycaemia may occur as a result of maternal administration.

DRUG INTERACTIONS

Administration of high doses of terbutaline prior to or shortly after anaesthesia with chloroform, cyclopropane, halothane, or trichloroethylene may increase the risk of severe ventricular arrhythmias, especially in patients with preexisting heart disease, because these anaesthetics greatly sensitize the myocardium to the effects of sympathomimetics. Enflurane, isoflurane and methoxyflurane may also cause some sensitization of the myocardium to the effects of sympathomimetics; caution is recommended during concurrent use with terbutaline. Tricyclic antidepressants and monoamine oxidase (MAO) inhibitors potentiate the action of terbutaline on the vascular system. Antihypertensive effects may be reduced when these medications are used concurrently with terbutaline. The patient should be carefully

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monitored to confirm that the desired effect is being obtained. Concurrent use of terbutaline with propranolol and other beta adrenoceptor blocking agents may result in mutual inhibition of therapeutic effects. Unwanted effects such as nervousness, irritability, insomnia or possibly convulsions or cardiac arrhythmia may result with the use of terbutaline and CNS stimulants. Hypokalaemia associated with high doses of terbutaline may result in increased susceptibility to digitalis induced cardiac arrhythmias. Caution and electrocardiographic monitoring are very important if concurrent use is necessary. Hypokalaemia may be enhanced by concomitant administration of aminophylline or other xanthines, corticosteroids or diuretic therapy. Possibility of cardiac arrhythmia is also increased with use of levodopa concomitantly. Concurrent use of nitrates with terbutaline may reduce the antianginal effects of nitrates. In addition to possibly increasing CNS stimulation, concurrent use with other sympathomimetics may increase the cardiovascular effects of either the sympathomimetic or terbutaline and the potential for side effects. Concomitant use of thyroid hormones with terbutaline may increase the effects of either these medications or terbutaline. Thyroid hormones also enhance the risk of coronary insufficiency when terbutaline is administered to patients with coronary artery disease.

OVERDOSAGE

In the alert patient who has taken excessive oral terbutaline, emesis should be induced followed by gastric lavage. In the unconscious patient, perform gastric lavage after the airway is secured with a cuffed endotracheal tube. Emesis never should be induced in an unconscious patient. Other supportive and symptomatic therapy include instillation of charcoal slurry to help reduce absorption of terbutaline, maintaining an adequate respiratory exchange and providing cardiac and respiratory support as needed. Patient should be monitored until he/she is symptom-free.

Up-to-date information on treatment of overdose can be obtained from The National Poison Centre, Universiti Sains Malaysia

TOXICOLOGY

Carcinogenicity/Tumorigenicity

An 18-month carcinogenicity study in NMRI strain mice showed that terbutaline administered orally in doses of 5, 50 and 200mg/kg (500, 5000 and 20,000 times the clinical subcutaneous dose, respectively) caused no drug related tumorigenicity. A 2- year carcinogenesis bioassay of terbutaline at doses of 50, 500, 1000 and 2000 mg/kg (corresponding to 167, 1667, 3333, and 6667 times the recommended daily adult oral dose, respectively) in Sprague-Dawley rats showed drug-related changes in the female genital system female rats showed drug-related increases in leiomyomas of the mesovarium, which were significant at doses of 500, 1000 and 2000 mg/kg; the incidence of ovarian cysts was increased significantly at all dose levels except at 2000 mg/kg; and hyperplasia of the mesovarium was increased significant at 500 and 2000mg/kg. However, in a 21 month study in mice, terbutaline at oral doses 5, 50 and 200mg/kg (corresponding to 17, 167 and 667 times the recommended daily adult oral dose, respectively) showed no evidence of carcinogenicity.

Mutagenicity

Long-term mutagenicity studies in animals have not been performed. There is no evidence from human data that terbutaline may be mutagenic.

Pregnancy/Reproduction

Fertility

A reproduction study in rats with oral terbutaline at doses up to 50mg/kg of body weight (mg/kg) (corresponding to 167 times the human oral dose) showed no adverse effects on fertility.

Pregnancy

Terbutaline crosses the placenta. Studies in humans have not been done. Parenteral administration of terbutaline during pregnancy has been reported to cause fetal tachycardia. Studies in animals (mice and rats) have not shown that terbutaline, at doses up to 1000 times the human dose, cause adverse effects on the foetus. Also reproduction studies in mice with terbutaline at doses up to 1.1 mg/kg, administered subcutaneously (corresponding to 4 times the human oral dose and 110 times the human subcutaneous dose) and in rats and rabbits with terbutaline at doses up to 50mg/kg orally (corresponding to 167 times the human oral dose and 5000 times the human subcutaneous dose) have shown no adverse effects on the foetus. (FDA pregnancy Category B)

Labour

Terbutaline inhibits uterine activity during the second and third trimesters of pregnancy and may inhibit labour. When administered during labour, terbutaline has been reported to cause serious adverse reactions such as transient hypokalaemia, pulmonary edema, and hypoglycaemia in the mother and hypoglycaemia in neonates of mother treated with parenteral terbutaline.

Breast-feeding

Problems in humans have not been documented. However, terbutaline is distributed into breast milk and some animal studies have shown that this medication to be potentially tumorigenic.

Paediatric

Appropriate studies on the relationship of age to the effects of terbutaline have not been performed in the pediatric population. However, no pediatric-specific problems have been documented to date.

Geriatric

No information is available on the relationship of age to the effects of terbutaline in geriatric patients.

Dental

No information was available.

Route of Administration: Oral

PRESENTATION

90ml

STORAGE CONDITIONS AND SHELF LIFE

Keep container tightly closed.
Store in a dry place below 30°C.
Protect from light.
Keep out of reach of children.
Jauhi daripada kanak-kanak.
Shelf life: please refer to outer package.

Product Registration Holder:
Duopharma Manufacturing (Bangi) Sdn. Bhd.
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